

Buck Converter Power Stage Design

Given Parameters

- Input Voltage: $V_{in} = 30 \text{ V}$
- Output Voltage: $V_{out} = 24 \text{ V}$
- Output Power: $P_{out} = 120 \text{ W}$
- Switching Frequency: $F_s = 100 \text{ kHz}$
- Ripple Current Rate: 30%
- Ripple Voltage Rate: 1%

Calculations

1. Duty Cycle (D)

$$D = \frac{V_{out}}{V_{in}} = \frac{24 \text{ V}}{30 \text{ V}} = 0.8 \quad (80\%)$$

2. Output Current (I_{out})

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{120 \text{ W}}{24 \text{ V}} = 5 \text{ A}$$

3. Output Resistance (R_{out})

$$R_{out} = \frac{V_{out}}{I_{out}} = \frac{24 \text{ V}}{5 \text{ A}} = 4.8 \Omega$$

4. Ripple Current (ΔI_L)

$$\Delta I_L = 0.3 \times I_{out} = 0.3 \times 5 \text{ A} = 1.5 \text{ A} \quad (\text{peak-to-peak})$$

$$I_{L_{\max}} = I_{out} + \frac{\Delta I_L}{2} = 5 \text{ A} + 0.75 \text{ A} = 5.75 \text{ A}$$

$$I_{L_{\min}} = I_{out} - \frac{\Delta I_L}{2} = 5 \text{ A} - 0.75 \text{ A} = 4.25 \text{ A}$$

5. Inductor (L)

$$L = \frac{V_{out} \cdot (V_{in} - V_{out})}{V_{in} \cdot F_s \cdot \Delta I_L} = \frac{24 \cdot (30 - 24)}{30 \cdot 100 \text{ kHz} \cdot 1.5} = 32 \mu\text{H}$$

Selection: 33 μH inductor with saturation current $> 5.75 \text{ A}$.

6. Ripple Voltage (ΔV_{out})

$$\Delta V_{out} = 1\% \times V_{out} = 0.24 \text{ V}$$

7. Output Capacitor (C)

$$C = \frac{\Delta I_L \cdot D}{8 \cdot F_s \cdot \Delta V_{out}} = \frac{1.5 \cdot 0.8}{8 \cdot 100 \text{ kHz} \cdot 0.24} = 6.25 \mu\text{F}$$

Selection: 10 μF , 35 V, low-ESR ceramic or polymer capacitor.

8. MOSFET Selection

- Voltage Rating: $> 30 \text{ V}$ (choose 50 V for margin)
- Current Rating: $> 5.75 \text{ A}$ (peak current)
- **Recommendation:** IRF540N (50 V, 22 A, low $R_{DS(on)}$)

9. Diode Selection

- Voltage Rating: $> 30 \text{ V}$ (choose 40 V for margin)
- Current Rating: $> 5.75 \text{ A}$ (peak current)
- **Recommendation:** SB1045 (40 V, 10 A Schottky diode)

Component Summary

Component	Value/Part Number
Inductor	33 μH , 7 A saturation
Capacitor	10 μF , 35 V, X7R/Ceramic
MOSFET	IRF540N (50 V, 22 A)
Diode	SB1045 (40 V, 10 A)